

Application of Quotient

2. FLT (6.27) = Fermat's Last Theorem

$x^n + y^n = z^n$ ($n \geq 3$) has no nontrivial solution.

RK1. Enough to show FLT is true for all $n \geq 3$

RK2 $x^p + y^p = z^p$ ($p \geq 3$) - Gauss - $\mathbb{Z}_p$

Recall. @ Partition

$$S = \bigcup_{i \in I} S_i \quad \left(\begin{array}{l} OS = \bigcup_{i \in I} S_i \\ OS_i \cap S_j = \emptyset \quad i \neq j \end{array} \right)$$

Equivalence relation

$$@ \quad \bar{s} = s / \sim \quad \forall s, t \in S, s \sim t \Leftrightarrow \exists i, s, t \in S_i$$

$$\text{if } S = V/U = \emptyset \quad s \sim t \Leftrightarrow s - t \in U \in \mathcal{L}(U) \Leftrightarrow s \in t + U$$

$$@ \quad \bar{v} = V/U = (\{v + U, v \in V\}, +, \cdot)$$

$$(v + U) + (v' + U) = (v + v') + U$$

$$(\lambda v + U) = \lambda v + U$$

Ex. $\mathbb{Z}_p = \mathbb{Z}/p\mathbb{Z} = (\{n + p\mathbb{Z} \mid n \in \mathbb{Z}\}, +, \cdot)$

$$= (\{0, 1, \dots, p-1\}, +, \cdot)$$

$$p \geq 3 \quad (a+b)^p = a^p + b^p$$

	0	1	2
0	0	1	2

	0	1	2
1	1	2	0

	0	1	2
2	2	0	1

$$\Leftrightarrow x^p + y^p = z^p \pmod{p}$$

Tanigawa-Shimura injection

Ex. $V = \mathbb{R}[x], U = (x^2 + 1)\mathbb{R}[x] = \{f(x) \in \mathbb{R}[x] \mid f(x) = (x^2 + 1)g(x)\}$

$$V/U = \{g(x) + U \mid g(x) \in \mathbb{R}[x]\}$$

$$g(x) + U = h(x) + U \Leftrightarrow h(x) - g(x) = (x^2 + 1)k(x)$$

$$\Leftrightarrow h(x) = ax + b + (x^2 + 1)k(x)$$

$$\dim V/U = 2$$

so $1 + U, x + U$ form a basis of V/U

$$@ \quad \cong \mathbb{R}[x]/(x^2 + 1)\mathbb{R}[x]$$

$$a + b\bar{x} \mapsto a + bx + U$$

$$(a + bx + U)(c + dx + U) = (a + bx)(c + dx) + U = (ac - bd) + (ad + bc)x + U$$

Cor. If $\dim V < \infty$, Then $\dim V/U = \dim V - \dim U$

Pf. Suppose $u_1, \dots, u_k$ is a basis of U

Expanding it to a basis of V $u_1, \dots, u_k, v_{k+1}, \dots, v_n$

claim: $v_{k+1} + U, v_{k+2} + U, \dots, v_{k+n} + U$ form a basis of V/U

Ex $R[x] / (x^2 + 1) R[x] \xrightarrow{\sim} \mathbb{C}$

$\forall f(x) \in \mathbb{R}^{T \times X}$ can define total orders

$$f(x) - g(x) \geq 0 \iff \underbrace{lc(f(x) - g(x))}_{\text{leading coefficient}} \geq 0$$

Cor. If $U \oplus T = V$ (i.e. T is a complement of U), then $V/U \cong T$

pf. Define $V/U \xrightarrow{\sim} T$ by $(U+U_i) \mapsto (U+U_i) + (U+U_i)$

$$u+t+U = v+U \implies \sigma(v+U) = \sigma(u+t+U) = t$$

- Key point: step 1 σ is well-defined

$$V + U = v' + U \xrightarrow{\sigma} \sigma(V + U) = \sigma(v' + U) = t$$
$$\downarrow \quad \quad \downarrow \quad \quad = \sigma(U' + U) = \sigma(U' + t' + U) = t'$$
$$t - t' \in U, \quad t - t' \in T$$
$$U \cap T = \emptyset \Rightarrow t - t' = 0 \Rightarrow t = t'$$

Step 2. $\sigma \in \text{Hom}_F(V/U, T)$ - mapping that

step 3. σ is an injection and a surjection

• $\mathcal{L}(V) = \frac{V}{\dots}$ $\dim V = 1$ $\dim V = 2$

$Fd_1 \oplus Fd_2 \oplus Fd_3$

$$F_2, \begin{matrix} \downarrow \\ \oplus \end{matrix} F_2, \begin{matrix} \downarrow \\ \dots \end{matrix} \dots \quad \begin{matrix} \downarrow \\ 0 \end{matrix} \quad F_2 \quad F_1 \quad \dots$$

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$a \cap b$

Thm (isom Thm of quotient)

$$\textcircled{1} a+b/a \cong b/a \cap b$$

$$\textcircled{2} a/a \cap b \cong a+b/b$$

$$\textcircled{3} a+b/a \cap b \cong a/a \cap b \oplus b/a \cap b \quad \text{over } \mathbb{Z} (v/a \cap b)$$

pf. $\textcircled{1}$ Define $a+b/a \xrightarrow{\sigma} b/a \cap b$

$$\text{by } \alpha + \beta + a \mapsto \beta + a \cap b$$

$$= \alpha' + \beta' + a \mapsto \beta' + a \cap b$$

$$\text{if } \alpha + \beta + a = \alpha' + \beta' \Rightarrow \beta - \beta' \in a \cap b$$

$$\textcircled{4} \alpha + \beta + a \cap b \mapsto (\alpha + a \cap b) + (\beta + a \cap b) \quad [3 \text{ steps}]$$